

5.13.12

AI25BTECH11036-SNEHAMRUDULA

Question:

The set of all values of λ for which the system of linear equations

$$2x_1 - 2x_2 + x_3 = \lambda x_1,$$

$$2x_1 - 3x_2 + 2x_3 = \lambda x_2,$$

$$-x_1 + 2x_2 = \lambda x_3,$$

has a non-trivial solution

- (a) contains two elements
- (b) contains more than two elements
- (c) is an empty set
- (d) is a singleton

Solution:

Consider the coefficient matrix

$$A = \begin{pmatrix} 2 & -2 & 1 \\ 2 & -3 & 2 \\ -1 & 2 & 0 \end{pmatrix}. \quad (0.1)$$

The characteristic equation is

$$\det(A - \lambda I) = 0. \quad (0.2)$$

On solving, the eigenvalues are obtained as

$$\lambda_1 = -3, \quad \lambda_2 = 1, \quad \lambda_3 = 1. \quad (0.3)$$

Corresponding eigenvectors are

$$v_1 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \end{pmatrix}, \quad v_3 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}. \quad (0.4)$$

Thus, the modal matrix is

$$P = \begin{pmatrix} -1 & 2 & -1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}. \quad (0.5)$$

The diagonal eigenvalue matrix is

$$D = \begin{pmatrix} -3 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}. \quad (0.6)$$

Inverse of P is

$$P^{-1} = \begin{pmatrix} \frac{1}{4} & -\frac{1}{2} & \frac{1}{4} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ -\frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{pmatrix}. \quad (0.7)$$

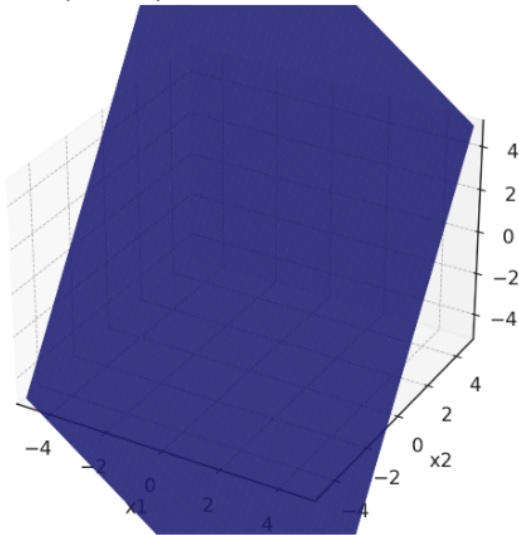
Hence, the eigendecomposition is

$$A = PDP^{-1}. \quad (0.8)$$

$$\{\lambda : \det(A - \lambda I) = 0\} = \{-3, 1\}$$

(A)Contains two elements.

3D plot of planes for $\lambda \approx 1.000000$



3D plot of planes for $\lambda \approx -3.000000$

